Welcome to ASDXDDL8M3- Data Modelling and Management

Moving around

Navigation

Right click + drag / Space bar + drag = Panning mode
+ / Mouse wheel = Zoom in
- / Mouse wheel = Zoom out
Ctrl + 2 = Zoom to selected object
Ctrl + 1 = 100% zoom
Ctrl + 0 = Adjust zoom to show all content
↑,↓,←,→ = Move on the board
Ctrl + Shift + [1-9,a-z] = Jump to quick link

Shortcuts to use

Selection

Click = Single object selection
Shift / CTRL + click = Multi selection
Shift + click & drag = Area selection
R = Toggle area selection
Ctrl + A = Select all
Alt + ↑,↓,←,→ = Select next object in the direction
Alt + Shift + ↓,↑ = Select group child or parent
Esc = Deselect object and reset to standard selection

Tools

N = Add sticky note
G = Add rectangular sticky note
S = Add shape
T = Create text
C = Add comment
L =Add line
D = Draw with nib brush
P = Previous brush in drawing mode
E = Eraser in drawing mode
Ctrl + L = Lock/unlock object
Ctrl + Shift + P = Pin/unpin object
Ctrl + H = Hide/unhide object

Have a play

Use the tools to get familiar with adding content.

	Add some post its!	Write something
Team 1		
Team 2		
Team 3		
Team 4		

Housekeeping

Rules regarding webcams
Keep your Webcam on throughout the delivery

Rules regarding mute
Keep your Microphone off throughout for privacy unless you are asked or need to respond.

Frequency of breaks
Short Breaks are scheduled every hour to support you with your learning.

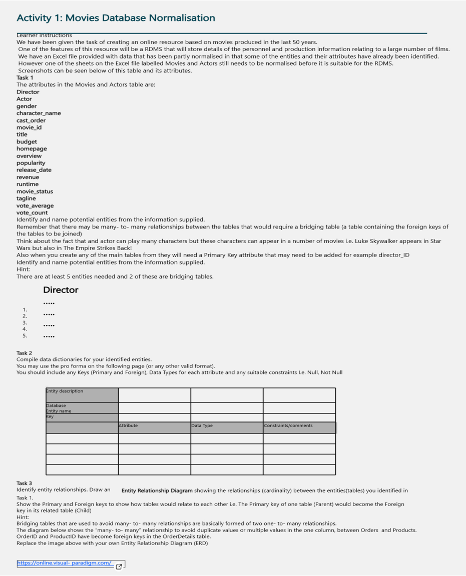
Rules regarding group participation
Everyone should equally participate in Group discission and take lead in contributing your view.

Rules regarding questions
The Golden Rule:
'There is no such thing as a stupid question'
'Even when asked by an instructor'
Please have a go at answering questions
Also:
'A question never resides in a single mind'
By asking questions, you are helping everyone out!

Anything else

Teams

Dominic	Shane	Toby					
Lucia	Ross	Sabrina	Stewart				
Bryan	Graeme	Paul	Sara				



Activity 1: Movies Database Normalisation

Learner instructions
We have been given the task of creating an online resource based on movies produced in the last 50 years.
One of the features of this resource will be a RDMS that will store details of the personnel and production information relating to a large number of films.
We have an Excel file provided with data that has been partly normalised in that some of the entities and their attributes have already been identified.
However one of the sheets on the Excel file labelled Movies and Actors still needs to be normalised before it is suitable for the RDMS.
Screenshots can be seen below of this table and its attributes.

Task 1
The attributes in the Movies and Actors table are:
Director
Actor
gender
character_name
cast_order
movie_id
title
budget
homepage
overview
popularity
release_date
revenue
runtime
movie_status
tagline
vote_average
vote_count
Identify and name potential entities from the information supplied.
Remember that there may be many- to- many relationships between the tables that would require a bridging table (a table containing the foreign keys of the tables to be joined)
Think about the fact that and actor can play many characters but these characters can appear in a number of movies i.e. Luke Skywalker appears in Star Wars but also in The Empire Strikes Back!
Also when you create any of the main tables from they will need a Primary Key attribute that may need to be added for example director_ID
Identify and name potential entities from the information supplied.
Hint:
There are at least 5 entities needed and 2 of these are bridging tables.

Director
.....
1. .....
2. .....
3. .....
4.
5. .....

Task 2
Compile data dictionaries for your identified entities.
You may use the pro forma on the following page (or any other valid format).
You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

Entity description			
Database Entity name			
Key			
	Attribute	Data Type	Constraints/comments

Task 3
Identify entity relationships. Draw an **Entity Relationship Diagram** showing the relationships (cardinality) between the entities(tables) you identified in Task 1.
Show the Primary and Foreign keys to show how tables would relate to each other i.e. The Primary key of one table (Parent) would become the Foreign key in its related table (Child)
Hint:
Bridging tables that are used to avoid many- to- many relationships are basically formed of two one- to- many relationships.
The diagram below shows the "many- to- many" relationship to avoid duplicate values or multiple values in the one column, between Orders and Products.
OrderID and ProductID have become foreign keys in the OrderDetails table.
Replace the image above with your own Entity Relationship Diagram (ERD)

https://online.visual- paradigm.com/

Task 1
Identify and name potential entities from the information supplied.

Identify and name potential entities from the information supplied.
Hint:
There are at least 5 entities needed and 2 of these are bridging tables.

1. Director
2. Actor
3. movies
4. MovieDirector (bridge)
5. ActorCharacterMovie (bridge)

Task 2
Compile data dictionaries for your identified entities.
You may use the pro forma on the following page (or any other valid format).
You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

Entity Description	Director Description		
Database Name	Movie and actors Database		
Entity Name	Director		
Key	Attribute	Data Type	Constrains (Null/Not Null)
Primary Key	Director ID	Int (6)	Not Null
	Director	varchar (50)	Not Null

Entity 2

Entity Description	Actor Description		
Database Name	Movies and Actors Database		
Entity Name	Actor		
Key	Attribute	Data Type	Constrains (Null/Not Null)
Primary Key	Actor ID	Int (6)	Not Null
	Actor	nvarchar (100)	Not Null
	Gender	varchar (20)	Null

Entity 3

Entity Description	Movies		
Database Name	Movies and Actors Database		
Entity Name	Movies		
Key	Attribute	Data Type	Constrains (Null/Not Null)
Primary Key	Movie ID	Int (6)	Not Null
	title	varchar (150)	Not Null
	budget	Money	Not Null
	Hompage	Varchar (MAX)	Null
	Movie Status	Varchar (20)	Not Null
	Overview	Varchar (200)	Not Null
	Release Date	Date	Not Null
	Revenue	Money	Not Null
	Runtime	Int (3)	Not Null
	Tagline	Varchar (500)	Null
	Vote average	Int (4,2)	Not Null
	Vote Count	Int (5)	Not Null
Foreign Key	Director ID	int (6)	Not Null

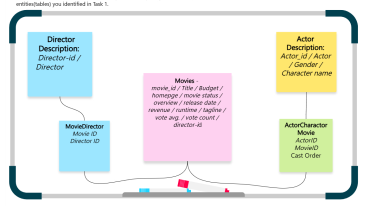
Entity 4

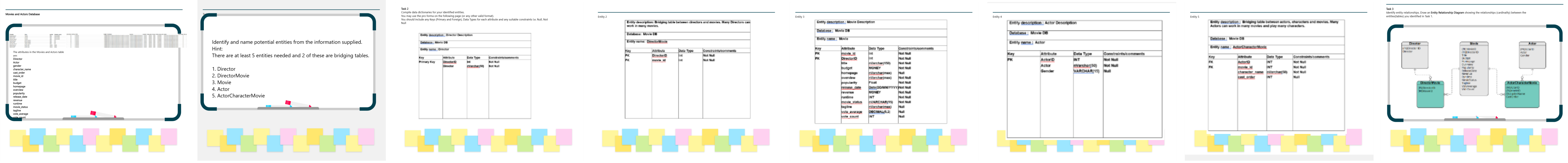
Entity Description	MovieDirector Bridge		
Database Name	Movies and Actors Database		
Entity Name	MovieDirector		
Key	Attribute	Data Type	Constrains (Null/Not Null)
Foreign Key	Movie ID	Int (6)	Not Null
Foreign Key	Director ID	Int (6)	Not Null

Entity 5

Entity Description	ActorCharacterMovie Bridge		
Database Name	Movies and Actors Database		
Entity Name	ActorCharacterMovie		
Key	Attribute	Data Type	Constrains (Null/Not Null)
	Actor ID	Int (6)	Not Null
	Movie ID	Int (6)	Not Null
	Cast Order	Int	Null
	CharacterName	Nvarchar(50)	Not Null

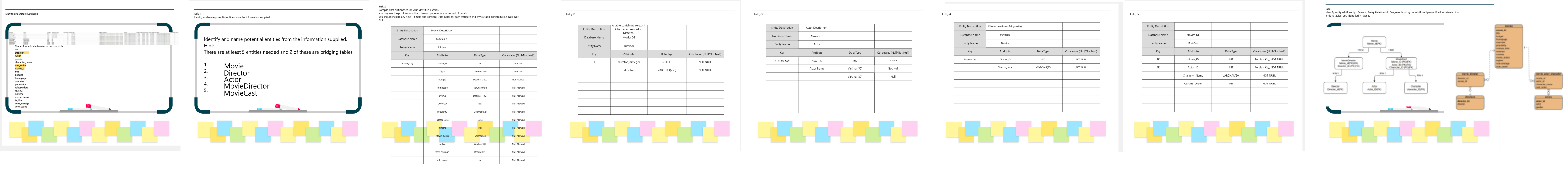
Task 3
Identify entity relationships. Draw an **Entity Relationship Diagram** showing the relationships (cardinality) between the entities(tables) you identified in Task 1.


Director Description: Director-id / Director
Actor Description: Actor_id / Actor / Gender / Character name
Movies - movie_id / Title / Budget / homepge / movie status / overview / release date / revenue / runtime / tagline / vote avg. / vote count / director-id
MovieDirector Movie ID Director ID
ActorCharactor Movie ActorID MovieID Cast Order



Identify and name potential entities from the information supplied.
Hint:
There are at least 5 entities needed and 2 of these are bridging tables.

1. Director
2. DirectorMovie
3. Movie
4. Actor
5. ActorCharacterMovie



Task 1
Identify and name potential entities from the information supplied.

Identify and name potential entities from the information supplied.
Hint:
There are at least 5 entities needed and 2 of these are bridging tables.

1. Movie
2. Director
3. Actor
4. MovieDirector
5. MovieCast